

2(a)(i)	<i>any two from:</i> 1 <i>ref. to</i> muscle fibre / sarcomere, contraction ; 2 active transport of (named), ion / molecule or active transport (of substance) against concentration gradient ; 3 movement of, vesicles / organelles, through cytoplasm / described ; 4 exocytosis of named substance ; 5 endocytosis / phagocytosis ; 6 spindle fibre / chromosome / chromatid, (movement) during, mitosis / meiosis ; 7 cilia / flagella, wafting / beating / AW ; I movement	2
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Question	Answer	Marks
2(a)(ii)	correct working ; e.g. $\frac{150}{0.2} = 750$ then $\frac{750}{24}$ or $\frac{150}{24} = 6.25$ then $\frac{6.25}{0.2}$ or $\frac{149.8}{0.2} = 749$ then $\frac{749}{24}$ 31 ;	2
2(a)(iii)	(respiration) oxidative phosphorylation ; (photosynthesis) <u>photophosphorylation</u> / light dependent stage ;	2
2(b)(i)	1 reduces ATP synthesis / less ADP reacts with P_i ; <i>any two from:</i> 2 protons <u>diffuse</u> through thermogenin ; 3 proton gradient reduced / reduces concentration of H⁺ in intermembrane space ; 4 fewer protons pass through ATP synthase ;	3
2(b)(ii)	<i>any three from:</i> 1 adrenaline binds to receptor on the cell surface membrane (of brown fat cells) ; 2 G-protein / adenylyl cyclase, activated ; 3 <i>ref. to</i> cAMP / second messenger, (is formed) ; 4 <i>ref. to</i> enzyme cascade / signal amplification / activation of kinase / signalling cascade ; 5 activation of lipase by phosphorylation ;	3